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Research Article

AI-assisted development of a rule-based pipeline for pre-examination screening in clinical neuropsychology: An exploratory method-development study

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Abstract

Objectives: We present an exploratory, method-development study of a framework for rule-based pre-examination screening. The study is exploratory: no clinical endpoint, success threshold, or control condition was pre-specified. Demonstrated on retrospectively analyzed irritable bowel syndrome (IBS) data, the pipeline produces a structured analysis with cross-domain pattern detection and audience-tailored reports. **Methods:** A large language model was used as a coding assistant to draft scoring logic, cross-domain rules, and Python code. The deployed pipeline executes these fixed rules deterministically; no model inference occurs when patient data are processed. The pipeline processes five instruments (Bergen Insomnia Scale, Conners' Continuous Performance Test, Chalder Fatigue Scale, Hospital Anxiety and Depression Scale, and the Repeatable Battery for the Assessment of Neuropsychological Status). For each patient, it generates an eight-step processing sequence, visualizations, and clinician-gated reports. Analyses used data from 105 adults (65 IBS, 40 controls) from the Bergen Brain–Gut–Microbiota study. **Results:** The tool flagged findings above threshold in sleep (50%), fatigue (40%), anxiety (35%), and attention (30%), with multi-domain flags in 60%. Cross-domain rules linked co-occurring findings to candidate mechanisms drawn from published literature. Two case illustrations demonstrated the rule-based summaries alongside clinician interpretation that went beyond the rules. **Conclusions:** The framework is a transparent support tool rather than an autonomous reasoning system: every output traces to a stated rule, and the contribution of artificial intelligence lies in development, not deployment: it augments rather than replaces clinical judgment. Because the analysis is exploratory and single-cohort, the findings are hypothesis-generating and require prospective, comparative validation.

Keywords: Decision support systems; clinical; diagnosis; computer-assisted; algorithms; reproducibility of results; large language models

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Statement of Research Significance

Research Question(s) or Topic(s): Can a large language model be used as a coding assistant to develop a transparent, rule-based system supporting neuropsychologists when planning an outpatient examination? **Main Findings:** Applied to 105 adults with and without irritable bowel syndrome, the system produced patient-level summaries highlighting clinically findings across sleep, fatigue, mood, and general cognitive status, with readable versions tailored for different audiences. **Study Contributions:** The main contribution of this exploratory study is methodological: it shows how a clinician can use a large language model while building a transparent screening tool that then runs on explicit rules alone, with no language model involved when a patient's data are processed. Because every output traces to a stated rule, the study offers a worked example of confining artificial intelligence to the design phase while keeping the clinical decisions with the neuropsychologist, who can revise the rules as evidence or local practice changes.

Introduction

In outpatient neuropsychological practice, clinicians often begin pre-examination preparation with a referral letter and intake questionnaires that must be synthesized into a clinically meaningful picture of the patient's difficulties, strengths, and needs. In practice this process is rarely so orderly: referral information is often incomplete and heterogeneous, and demand frequently exceeds capacity, with long waits and sustained pressure to improve throughput without sacrificing quality (DeRight & Puente, 2026).

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This synthesis means forming hypotheses about which domains matter most and which parts to prioritize, often under time pressure; relevant patterns may then be overlooked, and the examination may default to a standard battery rather than one tailored to the patient. Report writing itself remains a major workload burden (Postal et al., 2018). Tools that support efficiency while preserving clinical precision are therefore of interest.

This helps explain why artificial intelligence (AI) has become increasingly relevant to neuropsychology. Lundervold (2025) proposed a framework of precision neuropsychology in which AI presents both opportunities and risks: used carefully, it may reduce workload and extend care to underserved populations (Kronenberger et al., 2026), but must remain transparent, clinically grounded, and under professional oversight so that it strengthens rather than displaces clinical judgment (DeRight & Puente, 2026).

Despite advances in medical AI (Meskó & Topol, 2023; Thirunavukarasu et al., 2023), few concrete changes in everyday neuropsychological practice can be attributed to AI (DeRight & Puente, 2026); existing applications focus mainly on computerized testing, automated scoring, and diagnostic classification (Andersen et al., 2026; Parsons et al., 2018). These help organize scores but do not address the interpretive task at the center of the work: integrating cognitive, emotional, and environmental factors into a coherent formulation and the need to tailor reports to different audiences (Postal et al., 2018).

The present study presents a rule-based neuropsychological tool using a large language model (LLM) as a coding assistant (a conversational model used to generate and revise source code under human review). Our contribution lies in three areas: cross-domain integration and audience-tailored communication built on established cutoffs; a governance pattern in which a coding assistant helps specify rules that the finished tool executes deterministically at runtime; and case demonstrations of pre-examination screening before the first clinic visit.

Our framework is illustrated using a cohort enriched by patients with irritable bowel syndrome (IBS) (Berentsen et al., 2020). IBS is a common disorder of gut-brain interaction associated with abdominal pain, altered bowel habits, and reduced quality of life (Arif et al., 2025) and, through the gut–brain axis (Cryan et al., 2019; Mayer et al., 2015), with sleep disturbance (Tu et al., 2017), fatigue (Han & Yang, 2016), anxiety and depression (Staudacher et al., 2023), and subtler cognitive difficulties (Billing et al., 2024; Lam et al., 2019). Prior work found substantial profile overlap between IBS patients and controls (Lundervold et al., 2024, 2026); group-level cognitive differences were modest, so the development environment serves as a methodological testbed, with no claim that IBS is a priority population for neuropsychological evaluation.

We present a deterministic, rule-based pipeline for pre-examination screening in outpatient neuropsychological practice and report an exploratory, method-development demonstration using five established instruments. It applies published cutoffs and cohort-referenced rules to generate a structured overview, visual summaries, and written summaries for different audiences. The study is exploratively: no clinical endpoint, success threshold, or a priori hypothesis was pre-specified, there is no control condition, and the group comparisons below are secondary. Reporting guidance for AI-enabled tools (CONSORT-AI, DECIDE-AI; Liu et al., 2020; Vasey et al., 2022) was applied to the development process rather than to a deployed pipeline.

Method

Participants

For this study we used a convenience sample from the Bergen Brain–Gut–Microbiota (B-BGM) study, with no new participants or data acquired (Berentsen et al., 2020). The cohort comprises $N = 105$ adults: 65 patients meeting Rome IV criteria for IBS and 40 age- and gender-matched healthy controls (HC). Inclusion required an Irritable Bowel Syndrome Severity Scoring System (IBS-SSS) score ≥175, indicating at least moderate severity (Francis et al., 1997). Participants were adults (≥18 years); per the study protocol, those with severe or unstable somatic disease, known neurological disorder, or severe psychiatric disorder were ineligible. Controls were free of IBS or other disorders of gut-brain interaction. The study was conducted in accordance with the Declaration of Helsinki and approved by the Regional Committee for Medical and Health Research Ethics, South East Norway (REK2015-01621); all participants provided written informed consent.

The analysis-ready cohort file is a single participant-by-variable table of numerical scores from administered instruments, together with demographic variables and data-quality flags. It contains no free-text clinical material such as referral letters or journal notes. The table was derived from the source dataset after documented cleaning; the raw source file was preserved unchanged. "Data cleaning" here means a documented, scripted set of decisions: standardizing variable names and types, harmonizing missing-value codes, range-checking items and totals, and resolving or flagging inconsistent entries, not re-entry or re-scoring of test protocols. Unresolved values were flagged rather than corrected or imputed, and analyses were complete-case within each instrument (see the project repository).

Demographic characteristics are summarized in Table 1. The IBS group included three subtypes: IBS-D (diarrhea-predominant, $n = 24$), IBS-C (constipation-predominant, $n = 9$), and IBS-M (mixed, $n = 32$). IBS-SSS scores (available for $n = 56$) had a mean of $M = 273.8$ (SD = 71.3).

Neuropsychological instruments

Five domains were evaluated using established instruments, differing in administration requirements, which determine how much of the profile can be collected before the first clinic visit. Three self-report questionnaires (Bergen Insomnia Scale [BIS], Chalder Fatigue Scale, and Hospital Anxiety and Depression Scale [HADS]) can be completed remotely; the Conners' Continuous Performance Test, Third Edition (CPT-3) is self-administered but requires proprietary clinic software; and the Repeatable Battery for the Assessment of Neuropsychological Status (RBANS) requires a trained psychometrician. Thus, questionnaire data can precede the consultation while cognitive testing requires a clinic session. The scores include item-level responses for the three questionnaires, T-scores as exported by the CPT-3 software, and subtest raw scores for each of the single subtests of RBANS.

1. **Sleep/Insomnia – Bergen Insomnia Scale (BIS) [self-report]:** Six items scored 0–7 (days/week), total 0–42 (Pallesen et al., 2008). Pipeline cutoffs: Minimal (≤6), Mild (7–14), Moderate (15–24), Severe (≥25); clinical flags at total ≥19 or ≥3 items at ≥3 days/week.
2. **Sustained attention – Conners' CPT-3 [self-administered, clinic-based]:** A 14-minute continuous performance test

Table 1. Cohort demographics and neuropsychological domain scores by group

Measure	IBS ($n = 65$) *M (SD)*	HC ($n = 40$) *M (SD)*	t/χ^2	Cohen's d [95% CI]
Demographics				
Age (years)	37.8 (11.4)	35.6 (12.5)	0.90	0.19 [−0.21, 0.58]
Education (years)[a]	15.8 (1.9)	16.4 (1.4)	−1.54	−0.31 [−0.75, 0.14]
Female (%)	78.5	67.5	1.56†	0.12‡ [.00, .31]
Sleep[b]				
BIS total (0–42)	17.6 (7.6)	10.3 (6.9)	4.89***	0.99 [0.56, 1.42]
Attention (CPT-3 T-scores)[c]				
Detectability (d′)	48.9 (7.8)	44.3 (7.1)	3.00**	0.60 [0.19, 1.01]
Omissions	47.4 (6.4)	45.0 (1.6)	2.79**	0.45 [0.04, 0.86]
Commissions	51.0 (9.1)	48.0 (8.1)	1.70	0.34 [−0.07, 0.75]
HRT	48.4 (8.0)	48.7 (8.8)	−0.18	−0.04 [−0.44, 0.37]
Fatigue[d]				
Chalder total (0–11)	6.4 (3.4)	1.6 (2.5)	7.37***	1.55 [1.06, 2.04]
Emotional distress (HADS)[e]				
Anxiety (0–21)	8.1 (4.2)	4.2 (3.3)	4.97***	1.00 [0.56, 1.44]
Depression (0–21)	4.7 (3.1)	2.1 (2.3)	4.52***	0.90 [0.47, 1.34]
Neurocognition (RBANS index scores)[c]				
Immediate Memory	89.6 (16.1)	98.3 (15.7)	−2.65**	−0.54 [−0.95, −0.13]
Visuospatial	94.0 (11.7)	94.6 (12.4)	−0.24	−0.05 [−0.45, 0.35]
Language	98.2 (14.1)	100.9 (13.9)	−0.97	−0.20 [−0.60, 0.21]
Attention	92.5 (12.2)	100.0 (18.3)	−2.22*	−0.51 [−0.92, −0.10]
Delayed Memory	93.2 (15.0)	101.8 (19.8)	−2.30*	−0.51 [−0.92, −0.10]
Total Scale	90.5 (13.2)	99.2 (12.9)	−3.25**	−0.67 [−1.08, −0.25]

Note: †Pearson χ^2 test for gender (no continuity correction); all others are independent-samples Welch's *t*-tests. Effect size for the categorical sex comparison is Cramér's V; all other effect sizes are Cohen's *d*. Values in brackets are 95% confidence intervals: for Cohen's *d* from the large-sample (Hedges & Olkin) approximation, and for Cramér's V from the noncentral χ^2 distribution. *$p < .05$; **$p < .01$; ***$p < .001$. Sample sizes vary by measure due to missing data: [a]HC $n = 28$; [b]IBS $n = 58$, HC $n = 39$; [c]HC $n = 37$; [d]IBS $n = 49$, HC $n = 35$; [e]IBS $n = 57$, HC $n = 36$. Negative *d* values indicate IBS < HC. BIS = Bergen Insomnia Scale, CPT-3 = Conner's Continuous Performance Test, 3rd edition, HRT = hit reaction time, HADS = hospital anxiety and depression scale, RBANS = repeatable battery for the assessment of neuropsychological status.

yielding *T*-scored metrics (M = 50, SD = 10) including Detectability (d′), Omissions, Commissions, and Hit Reaction Time (Conners, 2014). Pipeline thresholds: $T \geq 60$ (mildly elevated), $T \geq 65$ (clinically significant).

3. **Fatigue – Chalder Fatigue Scale [self-report]:** Eleven items with bimodal scoring (0–11) covering physical and mental fatigue (Chalder et al., 1993). Pipeline caseness threshold: ≥4.
4. **Emotional distress – HADS [self-report]:** Anxiety (HADS-A) and Depression (HADS-D) subscales, each 0–21 (Zigmond & Snaith, 1983). Pipeline cutoffs: Normal (≤7), Borderline (8–10), Clinical (≥11) (Bjelland et al., 2002).
5. **General cognition – RBANS:** Ten subtests across five cognitive domains (Immediate Memory, Visuospatial/Constructional, Language, Attention, Delayed Memory), are administered in about 30 minutes using the Norwegian version (Randolph, 2013). For cohort-level comparisons (Table 1), conventional age-corrected index scores are used; within the pipeline, raw subtests are cohort-standardized and averaged into domain composites to keep the demonstration reproducible without redistributing proprietary norms. RBANS_Sum_Raw is used as a broad indicator of cognitive function. Cognitive flags require domain-summary or total percentile < 16. Percentiles are therefore cohort-referenced, not population-normed (Supplementary Material S1, E3).

The CPT-3 and RBANS are kept as separate domains because they assess complementary aspects of cognition: sustained attention and response inhibition versus broad cognitive function. This lets the rule-based engine distinguish patterns such as isolated, fatigue-driven attention deficits from broader cognitive involvement.

Two clarifications follow. First, although all five instruments are treated here as screening inputs, some, most clearly the RBANS, would normally be part of the formal examination itself. However, nothing in the present analysis depends on where the boundary falls, since all data were already available. In prospective use, each service needs to place the boundary for itself. Second, the questionnaires and clinic-based tests were obtained in the same period but not simultaneously; because fatigue, sleep, and mood are state-sensitive, this could affect symptom–performance alignment (noted as a limitation).

The deterministic pipeline

Architecture overview

The deterministic pipeline is a standalone Python program combining a quantitative analysis layer with a structured, rule-based interpretive layer. We use "processing sequence" in a restricted, procedural sense: it steps through a fixed protocol mirroring the sequence of a clinician's judgments but does not reason as generative models do. There is no inference engine, learning, or language model at runtime, only conditional application of implemented cutoffs and rules. It was applied to the analysis-ready cohort file; cleaning decisions requiring source verification were preserved as flagged values rather than overwritten, keeping interpretation conservative and auditable. Auditable means that a second reader can establish how an interpretation was reached: the inputs are recorded, the step from input to statement is stated, and the same inputs yield the same statement on any run. This is what allows a colleague to disagree on specific grounds rather than on impression, e.g., to see that a threshold is too low or high for a given population. Auditability is thus a property of the derivation, not of its correctness: a fully inspectable rule set can still use the wrong cutoffs.

Clarification on the role of AI. The pipeline was co-developed with the assistance of an AI coding assistant (Cursor IDE with

Table 2. Division of labor between the AI coding assistant, the clinician, and the deployed pipeline

Step	Performed by	Human verification
Development phase – AI-assisted		
Extraction of cut-offs, norms, and scoring algorithms from instrument manuals	AI coding assistant	Line-by-line comparison against the published manuals
Implementation of single-instrument scoring code	AI coding assistant	Code review; test cases with known input–output pairs
Proposal of cross-domain co-occurrence rules	AI coding assistant, within domains specified by the clinician	Clinical review; rules revised or rejected
Drafting of the chain and the audience-tailored output templates	AI coding assistant	Clinician revision of terminology and scope
Prompt log, model cards, and technical documentation	AI coding assistant and clinician	Completeness check against S1
Deployment phase – no AI		
Entry of raw instrument scores	Clinician or administrative staff	Automated range and completeness checks
Application of the approved rule set	Deterministic script	Not applicable; identical input yields identical output
Generation of the structured summaries	Deterministic template instantiation	Not applicable; wording is fixed
Interpretation, integration, and diagnostic conclusion	Clinician	Not applicable

Note: No model call is made during deployment, and no patient data is transmitted to any external service. All risk attributable to the AI coding assistant is confined to the development phase and is addressed by the verification step listed for each item.

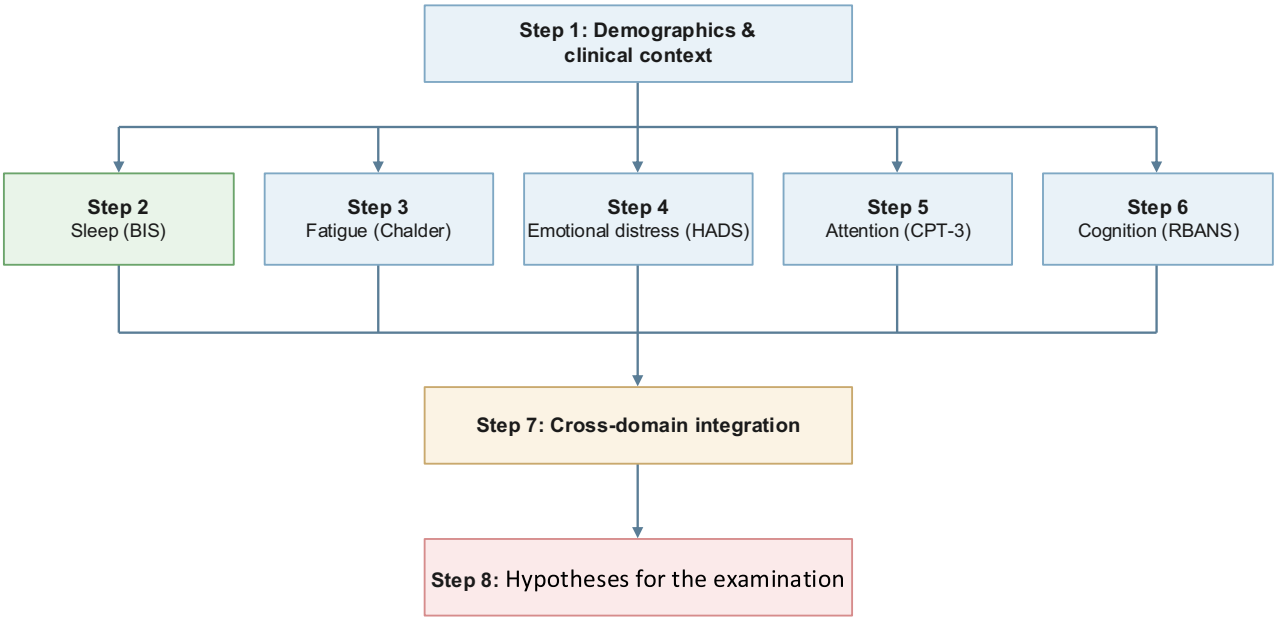


Figure 1. Eight-step processing sequence. Step 1 provides demographic and clinical context. Steps 2–6 assess five domains independently (parallel). Step 7 identifies co-occurring findings across domains. Step 8 produces hypotheses for the examination. The sequence mirrors the order in which a clinician would consider these findings, but the pipeline applies fixed rules at each step and does not reason. Interpretation remains with the clinician, and the outputs are inputs to further examination.

Claude Opus 4.6 and GPT-5.4) under human-in-the-loop review: every proposed rule, cutoff, and template was checked against published specifications before acceptance. When a patient's data are processed, the program applies published cutoffs and pre-authored templates locally, without network calls and with no generative AI involved (Table 3; Supplementary Material S1). We do not claim that the coding assistant adds computational capabilities beyond those achievable with conventional programming. Rather, we ask a feasibility question: can a clinician-led team build such a tool while keeping AI confined to the development? The deployed artifact contains no model. It is therefore deterministic, cannot generate factually unsupported content (termed "hallucination"), and transmit no data at run time. Its outputs depend instead on the quality of the rules. Processing followed ethics approval (REK2015-01621) and GDPR as implemented in Norwegian law; EU AI Act risk-tiering situates the tool as clinician-facing decision support (Supplementary Material S1, E4). The repository documents prompting strategy, design specifications, and model-card summaries.

Audience-tailored output design

A distinctive feature is the generation of outputs tailored to three audiences: clinical peers, referring physicians, and patients and families (Table 4). These differ in statistical detail, labelling, and visual language. All visualizations use a color-blind-friendly palette (Wong, 2011) and a common percentile-of-wellness scale (0–100, higher = better functioning), with inverted polarity where high raw scores indicate worse outcomes.

Recommendation engine

The pipeline identifies what to ask and what to test, leaving the conclusion to the examination itself. Each links to the justifying

Table 3. Audience-tailored output design: content and visualization *specifications*

Audience	Multi-panel figure	Radar plot
Clinical peers	Full statistical detail: *z*-scores, percentiles, effect sizes, confidence intervals. Technical labels.	Expanded 10-spoke version with Cohen's *d* inset.
Referring physicians	Traffic-light color coding (OK/Monitor/Action). Brief annotations for clinical triage.	Compact 5-spoke version. Traffic-light vertex coloring.
Patients & families	Accessible infographic. Plain-language labels, qualitative anchors, warm palette, no raw scores.	Compact 5 spokes with qualitative anchors and interpretation sentence.

Note: All variants display the same underlying data.

domain findings via conditional logic, referencing established guidelines where applicable (e.g., cognitive behavioral therapy for insomnia [CBT-I]: Grade A; Riemann et al., 2017). Because these outputs precede any examination, therapeutic items are provisional options for the clinician to weigh, not prescriptions, and are surfaced to patients or families only after clinician review; the system issues no advice autonomously.

Computational reproducibility

The pipeline source code and anonymized example outputs are available in a public GitHub repository (https://github.com/arvidl/AI-driven-neuropsych-screening). The cohort dataset is not public due to ethics conditions; dataset-level reproducibility is supported by documented cleaning decisions in the repository. The complete audience-tailored reports for the two case illustrations are provided as supplementary files.

Statistical analysis

Missing-data handling

Missingness here is predominantly blockwise (an entire instrument not administered e.g., the in-clinic CPT-3 or RBANS) rather than item-level. We therefore did not impute and analyzed each instrument complete-case, reporting the effective denominator for every measure. Because non-administration is plausibly unrelated to the unobserved scores, missing-at-random within domain is a reasonable working assumption (Schafer & Graham, 2002); we cannot exclude a missing-not-at-random component and therefore treat percentiles and comparisons as descriptive (Supplementary Material S1, E5).

Cohort-level group comparisons

IBS–HC differences across the five domains were reassessed on the cohort file as secondary, context-setting analyses. Comparisons used two-tailed Welch's t-tests with Cohen's d (pooled SD); the categorical sex comparison uses Cramér's V, at $\alpha = 0.05$ without correction for multiple comparisons, consistent with the exploratory aim; domain-specific complete-case samples were used. All effect sizes are reported with 95% confidence intervals (Table 1).

Pipeline output evaluation

Pipeline outputs were evaluated for computational accuracy (whether scores, percentiles, classifications, and flags matched the scoring rules), internal consistency after verification, and output completeness across the cohort. No clinical success threshold was pre-specified.

Results

Cohort characteristics

Demographic and clinical characteristics are presented in Table 1. The groups did not differ significantly in age ($t(77) = 0.90$, $p = .37$) or education ($t(68) = -1.54$, $p = .13$; HC $n = 28$). The cohort was predominantly female (74.3%), consistent with the sex distribution of IBS (Sperber et al., 2021), and the groups were otherwise well matched; domain-specific denominators vary with blockwise missingness.

Group differences across domains

Table 1 summarizes IBS–HC comparisons. Pronounced differences emerged for fatigue, anxiety, insomnia, and depression (all $p < .001$), with significant CPT Detectability and Omissions differences ($p < .01$). RBANS index scores differed for Total Scale and Immediate Memory ($p < .01$), Attention and Delayed Memory ($p < .05$), but not Visuospatial or Language.

These findings remain consistent with the gut-brain axis literature (Cryan et al., 2019; Mayer et al., 2015): IBS patients showed elevated insomnia, fatigue, and emotional distress relative to controls. These comparisons are secondary to the main purpose: demonstrating a transparent deterministic screening pipeline.

Pipeline output summary

The cohort file was processed successfully (MBP M4 Max) for all 105 participants in a single run (~500 s). Per participant, the pipeline generates an eight-step processing sequence (Table 2, Figure 1), a compiled PDF report, and visualization families (multi-panel profiles and radar plots) in three audience-tailored variants. In routine use the clinician consults a single clinician-facing summary; other variants demonstrate the architecture rather than proposing that all outputs enter the pre-examination workflow.

Distribution of clinical findings

Domain-specific flags across the 105 pipeline summaries (criteria as in the Method) were distributed as follows (IBS subset in parentheses): sleep (BIS) 53, 50% (40); neurocognition (RBANS) 48, 46% (36); fatigue (Chalder) 42, 40% (36); anxiety (HADS-A) 37, 35% (31) and depression (HADS-D) 8, 8% (7); attention (CPT-3) 32, 30% (24). Multi-domain elevations (flags in ≥2 domains) occurred in 63 participants (60%), predominantly IBS ($n = 51$).

Case illustrations

We present two illustrative cases. Table 2 summarizes the eight-step processing sequence across both and contrasts a circumscribed sleep-centered profile with a broader multi-domain profile. The chain is deliberately open at this end: it identifies what to ask and what to test, leaving the conclusion to the examination itself.

Table 4. Comparative summary of the pipeline's eight-step processing sequence across the two illustrative cases, ending in hypotheses for the examination rather than diagnostic or prognostic conclusions

Step	Case 1: Focal sleep problems	Case 2: Multi-domain involvement
1. Context	Male, 38 yrs, 12 yrs education, IBS-D, moderate IBS severity.	Female, 31 yrs, 16 yrs education, IBS-M, high IBS severity.
2. Sleep	BIS = 16 (Moderate; 59th cohort percentile). Clinically elevated, with four items endorsed at ≥3 days/week.	BIS = 32 (98th cohort percentile). Severe insomnia and highly elevated sleep-domain burden.
3. Fatigue	Chalder = 3 (below caseness threshold). No fatigue flag.	Chalder = 11 (97th cohort percentile). Maximum fatigue caseness.
4. Mood	HADS-A = 2 and HADS-D = 2 (both Normal). Emotional distress not flagged.	HADS-A = 12 (clinical range), HADS-D = 6 (Normal). Anxiety flagged without depressive caseness.
5. Attention	CPT-3 Detectability $T = 60$ (mildly elevated), with remaining CPT metrics within normal limits.	CPT-3 profile within expected range; sustained attention remained comparatively preserved.
6. Cognition	RBANS Sum Raw = 233 (57th cohort percentile). Broadly mid-range profile; Word List approached the lower boundary of normal range.[a]	RBANS Sum Raw = 213 (15th cohort percentile). Marked relative weakness in Language, with additional weakness in Immediate Memory, Delayed Memory, and Recognition.[a]
7. Cross-domain integration	Flags span sleep and attention, supporting an interacting symptom pattern with potential sleep-related attentional inefficiency.	Sleep, fatigue, anxiety, and cognition form a multi-domain pattern. The pipeline links insomnia and fatigue, notes the bidirectional mood-sleep relationship, and identifies anxiety as a contributor to cognitive inefficiency.
8. Hypotheses for the examination	Whether the flagged sleep difficulties account for the reported daytime symptoms, and whether they are amenable to targeted follow-up, are questions for the examination to address.	Mood findings co-occur with lower language and memory scores. Test whether these are secondary to emotional distress, or whether the language findings are independent.

Note: [a]RBANS scores are cohort-relative rather than population-normed.
BIS = Bergen Insomnia Scale, CPT-3 = conners' continuous performance test, 3rd edition, HADS = hospital anxiety and depression scale, HADS-A = HADS anxiety subscale, HADS-D = HADS depression subscale, IBS = irritable bowel syndrome, IBS-D = diarrhea-predominant IBS, IBS-M = mixed IBS, RBANS = repeatable battery for the assessment of neuropsychological status.

Case 1: IBS patient with circumscribed sleep disturbance

The patient is a 38-year-old male with IBS-D (moderate severity, IBS-SSS = 195).

Pipeline output

As summarized in Table 2, the pipeline produced a focal profile centered on elevated sleep disturbance, with Detectability at the CPT flag threshold and otherwise preserved fatigue, mood, and cohort-relative cognition. The processing sequence flagged sleep and sustained attention as priority domains for examination, noting that their relationship requires clarification: it suggested taking a detailed sleep history and assessing whether the attentional finding is better accounted for by sleep disturbance or by another mechanism. As with all pipeline outputs, these are starting points for clinician review, not prescriptions.

The corresponding clinical multi-panel profile and audience-tailored radar plots are shown in Figures 2 and 3. The complete report for this case, comprising the clinician and physician versions, is provided as Supplementary Material S2 (case_1_-report.pdf). A patient-facing rendering of the same findings is included to show that a single rule set can express one set of results at three levels of technical detail without recomputing. Its place in the workflow will be commented on in the Discussion: communicating findings to a patient before an examination has taken place is not something we propose.

Clinical interpretation of Case 1 after reading the rule-based report

Beyond the rule-based report, the clinician treated the sleep-centered reading as provisional. Reduced Detectability (d′) is a particular concern: in IBS, hypervigilance toward somatic and threat-related stimuli may lower the response criterion independently of sleep quality, reducing the d′ without a primary attentional capacity deficit (Akbari et al., 2025). If low d′ reflects hypervigilance rather than attentional capacity, sleep-focused intervention alone may miss the target. The examination should therefore disentangle these pathways and determine whether indications of verbal memory weakness reflect sleep-mediated consolidation failure or a more specific cognitive vulnerability before deferring broader follow-up. This distinction lies outside what screening scores can settle: no rule applied to these variables can separate a more cautious response threshold from reduced attentional capacity, since the two produce the same pattern at this level of measurement. The case therefore illustrates where rule-based integration ends and examination begins. Whether such ambiguities recur systematically enough to warrant a rule that flags them is a question for prospective work.

Case 2: IBS patient with multi-domain involvement

A second IBS patient (IBS-M subtype, female, age 31, IBS-SSS = 305) provides a contrasting example of multi-domain involvement.

Pipeline output

As summarized in Table 2, the second case showed a broader profile with severe insomnia, maximum fatigue caseness, clinical anxiety, and cohort-relative RBANS weakness, while sustained attention and depression were unflagged. Cross-domain rules linked insomnia and fatigue, noted the mood–sleep relationship, and flagged sleep, fatigue, and anxiety as priority domains, with cognitive weakness flagged for clarification as secondary or independent. The high IBS symptom burden was noted as relevant context for the examination and for coordination with the treating gastroenterologist. As with all pipeline outputs, these are starting points for clinician review, not prescriptions.

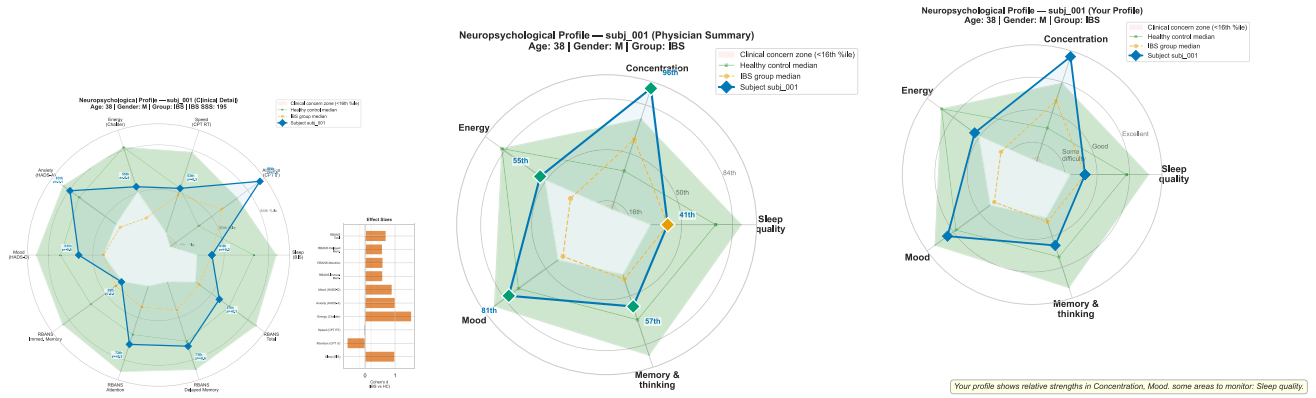
The multi-panel and radar visualizations for this case are shown in Figures 4 and 5. The complete report for this case is provided as Supplementary Material S3 (case_2_report.pdf), constructed as for Case 1.

Clinical interpretation of Case 2 after reading the rule-based report

Beyond the rule-based report, clinician interpretation noted a profile specificity the rules did not capture. The severe Naming deficit is more consistent with a language-network vulnerability



Figure 2. Multi-panel domain profile (clinical version) for Case 1. Each panel shows the cohort distribution (violin plots split by IBS/HC group) with the patient's score marked as a blue diamond. Panel 1 (BIS): Insomnia total with clinical threshold. Panel 2 (CPT): Attention metrics with T-scores exported by the test software. Panel 3 (Chalder): Fatigue with caseness threshold. Panel 4 (HADS): Anxiety and Depression with classification zones. Panel 5 (RBANS): Raw-subtest-derived summary composites and cohort-referenced position. Illustrative pipeline output; the visualizations are intended to support rather than replace clinical judgement.



(a) Clinical peers (b) Referring physicians (c) Patients & families

Figure 3. Audience-tailored radar plots for Case 1. All three variants display the same values on a common normalized scale, oriented so that higher positions indicate better functioning, and differ in level of detail and visual language. (a) The clinical version uses an expanded 10-spoke layout with *z*-score annotations and a Cohen's d inset. (b) The physician version uses a compact 5-spoke layout with traffic-light coloring. (c) The patient version uses "plain language labels", qualitative anchors, and a warm palette, and additionally carries a summary sentence describing the profile. Illustrative pipeline output; the visualizations are intended to support rather than replace clinical judgement.

than with sleep disruption or fatigue, which tend to impair processing speed and retrieval but not confrontation naming. This, together with a dissociation between near-average attention and severely reduced memory performance, argues against a unitary sleep–fatigue explanation. The clinician therefore raised the possibility of a primary cognitive vulnerability and judged immediate rather than deferred neuropsychological follow-up. Unlike Case 1, the screening data support this distinction: the pipeline captured the subtest-level pattern underlying the clinician's interpretation. The rules lacked explicit knowledge that confrontation naming is comparatively spared by sleep disruption and fatigue. A rule encoding that asymmetry could be specified. Cases of this kind indicate where the rule set could be extended.

Discussion

We demonstrated a deterministic, rule-based pipeline for pre-examination screening on 105 participants (65 IBS, 40 HC). It flagged elevation above threshold across five domains, applied cross-domain rules, and produced audience-tailored outputs for clinical

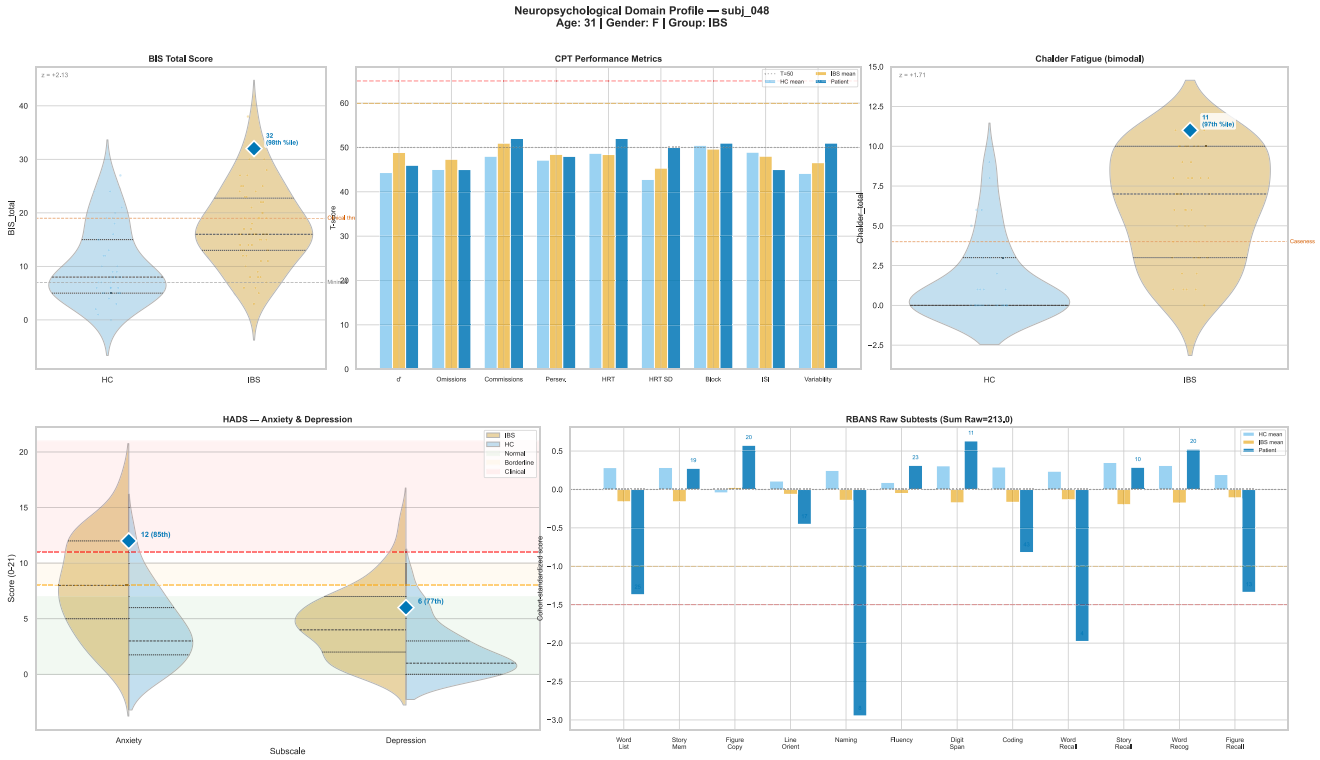
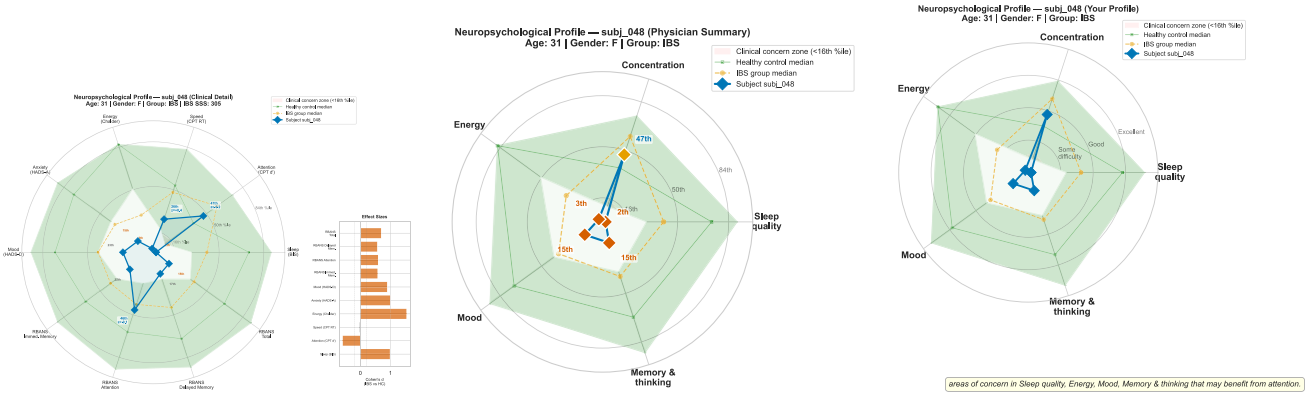


Figure 4. Multi-panel domain profile (clinical version) for Case 2. Panels are constructed as in Figure 2; each shows the cohort distribution as violin plots split by group, with the patient's score marked as a blue diamond, together with the relevant thresholds. Illustrative pipeline output; the visualizations are intended to support rather than replace clinical judgement.



(a) Clinical peers (b) Referring physicians (c) Patients & families

Figure 5. Audience-tailored radar plots for Case 2. Variants are constructed as in Figure 3. Illustrative pipeline output; the visualizations are intended to support rather than replace clinical judgement.

peers and physicians; a patient-facing variant is generated but released only after the clinical evaluation. The tool provides the neuropsychologist with a structured synthesis before the encounter, intended to support more targeted interview and test selection. Its contribution is methodological and integrative rather than diagnostic. It applies stated rules to combine findings across domains and presents the results in a form suited to the end user. This is not a clinical formulation in the full professional sense. Weighing competing differential hypotheses and deciding between them, as both case illustrations show, remain a clinical responsibility.

Relation to prior work

Traditional template-based systems can populate summaries with scores but typically provide more limited support for cross-domain aggregation. A central feature of the present pipeline is the eight-step processing sequence, which applies its rules in a fixed sequence corresponding to the domains a clinician would review. Rather than listing isolated elevations, it links findings across domains and generates candidate hypotheses about how sleep, fatigue, emotional distress, attention, and cognition may interact, drawn from relationships reported in the published literature.

The deterministic architecture matters: clinicians need to understand why a tool produced a given output or interpretation and to challenge it when warranted. Because every classification, flag, step, and suggested line of inquiry traces to an explicit rule, the pipeline offers transparency, reproducibility, and traceability. These properties are not incidental: in a domain where interpretive judgment is a core professional competency, opaque generative output would be difficult to justify clinically or professionally. Another strength is related to the fact that a full cohort of 105 participants could be processed and corresponding reports generated in a single, quick run.

Clinical implications

From a workflow perspective, the pipeline is a structured briefing tool. Many outpatient services already collect questionnaires before the appointment (e.g., King's College Hospital NHS Foundation Trust, 2026; Stanford Health Care, 2026). The framework extends this by integrating the screening data before the first encounter into a cross-domain summary, flagged areas, possible interactions, and priority domains and open questions for the examination.

The summary may help prioritize patients with multi-domain elevations, guide follow-up instrument selection, and serve as a discussion aid. Because the same cutoffs apply across patients, it may reduce inconsistency from idiosyncratic review and provide a starting point for documentation (Postal et al., 2018). The cross-domain component does not infer causality but highlights co-occurring findings that might otherwise remain fragmented across instruments; these outputs are hypothesis-generating and require clinical verification.

Two cautions about the inputs deserve emphasis. First, these instruments are screeners, not diagnostic tools: elevated anxiety, fatigue, or mood scores are non-specific. Elevated fatigue, for instance, is equally consistent with disturbed sleep, a mood disorder, or the somatic condition itself, and differential diagnosis remains a clinical task. Second, the efficiency the framework offers and the risk it carries arise from the same property. A pre-examination summary saves preparation time because it arrives already organized, but it may make clinicians susceptible to anchoring on an initial formulation and to seeking information that confirms it. A fluent, structured output presented before the examination may thus narrow the questions asked and the measures selected, thereby reduce the chance that such errors being detected. Case 1 illustrates both sides: the tool correctly identified sleep and attention as the salient domains, while the clinician's consideration of somatic hypervigilance as an alternative account of reduced d′ came from clinical knowledge the tool does not hold.

The risk therefore constrains how the outputs are phrased, not only what they contain: a summary that poses a question anchors less than one that states a conclusion, even where the underlying information is identical. Confining the tool to priority domains, competing explanations, and suggested lines of inquiry keeps the outputs interrogative rather than declarative, which preserves the preparatory benefit while leaving interpretation to the clinician at the point where independent judgment matters most. This does not eliminate anchoring. We regard the residual risk as a limitation requiring empirical study rather than one resolved by design.

A further constructive direction, arguably more valuable than presenting findings, is to invert the emphasis so the system also surfaces what is missing or underweighted: differential diagnoses the screeners cannot exclude, internal inconsistencies, and data the pipeline did not receive, thereby challenging rather than reinforcing the clinician's initial impression (Supplementary Material S1, E6).

The audience-tailored outputs address differing stakeholder needs (Gruters et al., 2022; Postal & Armstrong, 2024; Postal et al., 2018): brief, flagged summaries for referring physicians; full detail for clinical peers. Traffic-light categories are screening flags, not findings. Patient- and family-facing variants are generated from the same rule set but withheld until after evaluation; whether they improve understanding requires separate study.

The case illustrations clarify a boundary condition: the pipeline identifies priority domains and candidate interactions, but the clinician supplied interpretations the rules could not: hypervigilance versus capacity (Case 1) and naming-specific vulnerability (Case 2). All outputs require professional review and may need modification when language, education, culture, or referral context alters the data's meaning (Lundervold, 2025).

Limitations

This is a single-cohort, cross-sectional exploratory demonstration on a specific battery (BIS, CPT-3, Chalder, HADS, RBANS). Extension to other batteries or populations would require reconfiguring cutoffs, mappings, and domain logic. The limitations fall into two groups: those inherent to a rule-based approach, which would persist in any dataset, and those specific to the present demonstration, which a prospectively designed study would remove.

Limitations of the approach. The cross-domain rules encode assumed relationships between constructs, for example between disturbed sleep and memory consolidation, drawn from published literature but untested in this cohort, and established co-occurrence rather than causality. The pipeline receives questionnaire item scores and test-derived CPT-3 and RBANS values, but no free-text history, interview, or mental-state examination. Cutoffs are applied uniformly, which aids consistency but does not adjust for age, education, language, culture, sensory or motor status, or comorbidity. The pipeline flags incomplete domains and adjusts denominators but cannot represent why data are absent. A discontinued or omitted test is in itself a finding. Clinical use would therefore require that reasons for non-administration be captured as structured input and treated by the rule set as informative rather than missing.

Limitations of the present demonstration. The cohort was a convenience sample of de-identified research data. Validation here concerns computational accuracy (i.e., agreement between the pipeline's output and its own specification), rather than clinical validity. Comparison against independent clinical interpretation is not available for this cohort. Reasons for non-completion are likewise unavailable, so group comparisons used complete-case analysis under a working MAR assumption and are descriptive only. If incomplete cases differ systematically from complete ones, which cannot be assessed here, profiles may be drawn from a non-representative subset. Cohort-referenced percentiles rest on $N = 105$. Measures were obtained within the same period but not simultaneously, and the full pre-examination workflow was not achievable because the dataset required clinic-based CPT-3 and RBANS collection. Outputs were not co-designed with patients, families, or physicians, and stability and accessibility were not examined.

Future directions

Future work should address clinical translation and further development. One direction pairs the framework with cognitive tools that allow more of the profile to be collected before the first visit; another is longitudinal application; monitoring symptom change and treatment response. Its modular architecture also supports extension to additional instruments and populations.

A priority is a prospective, comparative evaluation against a clinical reference standard, contrasting usual pre-examination planning with pipeline-supported planning using pre-specified endpoints: preparation time, agreement of plans and conclusions with expert consensus or eventual diagnosis, error and omission rates, and induced biases such as anchoring. A prospective collection is required rather than reanalysis of existing data, since the clinical record is not recoverable from a de-identified cohort. The same design could support the psychometric work the rule set requires; convergent validity of the screening instruments against established measures in the same domain, and comparison of the tool's domain classification against diagnoses confirmed by recognized standards. A complementary line is to build and evaluate a structured disconfirmation pass that surfaces alternative interpretations and counteracts anchoring on the initial output.

Equally important are usability and implementation studies, with neuropsychologists, physicians, and patients, determining whether audience-tailored outputs improve planning, communication, and efficiency. We regard stakeholder co-design as an outcome in its own right. Patient-, family-, and physician-facing outputs require separate development and evaluation as a post-evaluation communication tool, including comprehensibility and acceptability testing, before use - across varied language backgrounds, digital access levels, health literacy, and disability-related needs.

It should also address how the workflow accommodates individuals who decline any involvement of artificial intelligence in their care. The distinction between a language model used to build the tool and one running inside it is technically meaningful, but whether patients regard development-phase involvement as material to consent is an empirical question rather than one settled by design. In our case, no language model processes patient data, and the deployed tool can be inspected rule by rule; whether that suffices for a given patient is for that patient to judge and should be asked directly in the implementation studies proposed above.

In the longer term, integration with electronic health records may support routine use, provided transparency, governance, and accountability remain central. Recent work on characterizing AI capabilities emphasizes evaluating each faculty robustly against a meaningful human baseline (Burnell et al., 2026). The same principle applies here: the framework should be validated component by component rather than as a single undifferentiated tool. The relevant components are the accuracy of each domain classification, the soundness of each cross-domain rule, and the effect of the outputs on clinician planning. Because the rules are stated explicitly rather than learned, a service can inspect them, adapt cutoffs or add domains to its own population and referral pattern, and re-verify the result. This is the practical form the contribution takes.

Conclusions

This exploratory demonstration shows that a clinician-led team can use an AI coding assistant to build a transparent, auditable pre-examination screening tool that uses no AI at the point of care. Within the limits of the dataset and rule set, the pipeline identified clinically relevant flags, linked co-occurring findings, and produced audience-tailored outputs. Formal comparative validation, usability testing, and workflow integration remain prerequisites for clinical use.

Supplementary material. The supplementary material for this article can be found at https://doi.org/10.1017/S1355617726102276.

Data availability statement. The cohort dataset contains clinical neuropsychological data collected under ethics approval and is not publicly available due to participant confidentiality and ethics approval conditions. Data may be made available upon reasonable request to the corresponding author, subject to institutional data sharing agreements. The pipeline source code and anonymized example outputs are openly available at (https://github.com/arvidl/AI-driven-neuropsych-screening). The complete audience-tailored reports for the two case illustrations are provided as Supplementary Material S2 and S3.

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Competing interests. The authors declare none.

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